
SYSTEMATIC LONG-SHORT INVESTMENT STRATEGY

Risk-Adjusted Momentum with Utility-Based Optimization

Group 13

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Investment Portfolio Management | Master 1
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Investor Profile	Very Aggressive (BRTI = 6.6)
Risk Parameters	$\lambda = 1.76$ $q = 0.93$
Strategy Type	Long-Short Market-Neutral
Gross Leverage	200%
Live Performance	+36.25% (Feb–May 2026)

ACADEMIC INTEGRITY STATEMENT
LLMs assisted with coding only;
report text and interpretations are the authors' own work.

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1. Risk Profile & Investment Policy Statement

1.1 Behavioral Risk Questionnaire

We completed the BRTI questionnaire (10 questions, scale 1–7; questions 2, 3, 8, 9 reverse-scored: $Q^* = 8 - Q$; full scoring in Appendix A). Our derived parameters are:

$$q = \frac{\text{BRTI} - 1}{6} = \frac{5.6}{6} = \mathbf{0.93} \quad (1)$$

$$\lambda = 12.5 e^{-0.35(\text{BRTI}-1)} = 12.5 e^{-1.96} = \mathbf{1.76} \quad (2)$$

This places us in a **Very Aggressive** profile (top of the BRTI scale). Our indifference curves (Appendix A, Fig. 6) are nearly flat, confirming a low variance penalty and our willingness to hold concentrated positions.

Implications for our portfolio construction. Our low $\lambda = 1.76$ has three direct consequences. (i) *Target volatility*: our optimizer accepts high net volatility; we do not cap portfolio volatility, which justifies the 200% gross leverage we employ. (ii) *Aggressiveness of long-short exposures*: our position weights concentrate at the [1%, 25%] bounds rather than spread evenly, amplifying the momentum signal. (iii) *Risk controls*: because our utility function provides only a mild variance penalty, we impose explicit structural guards, specifically the 30% per-sector cap and the minimum 10L/10S constraint, to prevent extreme concentration in a single sector or factor.

1.2 Investment Constraints

Table 1: Strategy constraints

Parameter	Value
Universe	S&P 500 current constituents
Gross leverage	200% (100% long + 100% short)
Min. positions	10 long + 10 short
Sector cap	30% per GICS sector per leg
Transaction cost	USD 2/trade Margin: 8%/yr on short book
Risk-free rate	$R_f = 2\%$ per annum

2. Strategy Design

2.1 Signal: Risk-Adjusted Momentum

$$\text{Score}_i = \frac{R_i^{(12m-1m)}}{\hat{\sigma}_i^{(60d)}} \quad (3)$$

$R_i^{(12m-1m)}$ is the 12-month return excluding the last month (to avoid short-term reversal); $\hat{\sigma}_i^{(60d)}$ is the 60-day realized annualized volatility. We assign the top 10 scores to the **long leg** and the bottom 10 to the **short leg**.

2.2 Optimization & Efficient Frontier

$$\max_w U = \mathbb{E}[R_p] - \frac{\lambda}{2}(\sigma_p^2)^q \quad (4)$$

$\lambda = 1.76$, $q = 0.93$ (from our BRTI). We optimize each leg separately via SLSQP; weights $\in [1\%, 25\%]$; 30% sector cap. When $q = 1$, this reduces to standard Markowitz.

Expected return and covariance estimation. At each monthly rebalancing date, we compute expected returns as the annualized mean of daily returns over the past 60 trading days ($\hat{\mu}_i = \bar{r}_{i,60d} \times 252$), and we estimate the covariance matrix from the same 60-day window of daily returns, scaled to annual frequency ($\hat{\Sigma} = \Sigma_{60d} \times 252$). This short estimation window keeps our inputs responsive to the current market regime and is consistent with the 60-day volatility denominator used in our signal.

Figure 1 displays our mathematical efficient frontier computed over our 20-stock selection universe (10 long candidates, 10 short candidates) subject to the 30% sector cap. Each point on the navy curve represents the minimum-variance portfolio achievable for a given target return level, obtained by solving 120 constrained quadratic programs.

Three observations are worth noting. First, our optimal portfolio (red $\star$) lies close to but not exactly on

the frontier. This is a direct consequence of our construction: we optimize the long and short legs *separately* using Eq. (4), whereas the frontier is computed over all 20 stocks jointly, the two problems are not equivalent. Second, our portfolio sits relatively far to the left (low net volatility) despite the 200% gross leverage. This reflects the hedging property of our long-short structure: the negative correlation between the two legs reduces our net portfolio variance well below what the individual legs would suggest. Being “Very Aggressive” therefore translates into high concentration and large position sizes for us, not necessarily into high net volatility. Third, our indifference curve (red dash-dot) passes through the optimal portfolio and is nearly flat, confirming that our calibrated $\lambda = 1.76$ imposes only a mild variance penalty, we are willing to accept substantial additional risk for modest incremental return.

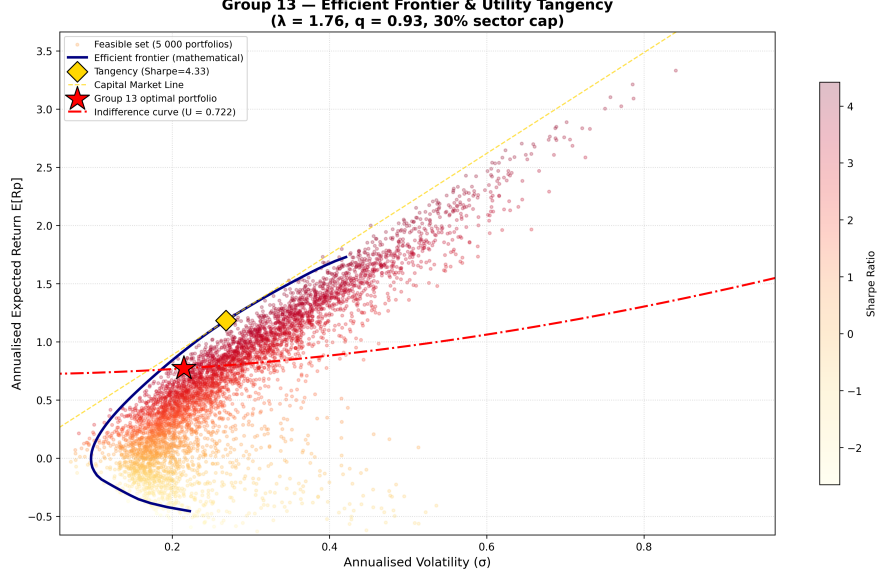


Figure 1: Efficient frontier (navy), Group 13 optimal portfolio (red $\star$), tangency portfolio (gold $\diamond$), CML, and indifference curve of Eq. (4) passing through the optimum.

3. Backtest & Live Implementation

3.1 Backtest Design (Jan 2016 – Dec 2024)

We rebalance monthly with no look-ahead bias. Monthly costs: $20 \times \$2 = \40 (fees) + $\$1\text{M} \times 8\% / 12 \approx \$6,667$ (margin). Starting capital: USD 1,000,000. We download daily closing prices via `yfinance` for all S&P 500 constituents (474 stocks retained after delisting filter); sample: 2015-01-01 to 2024-12-31.

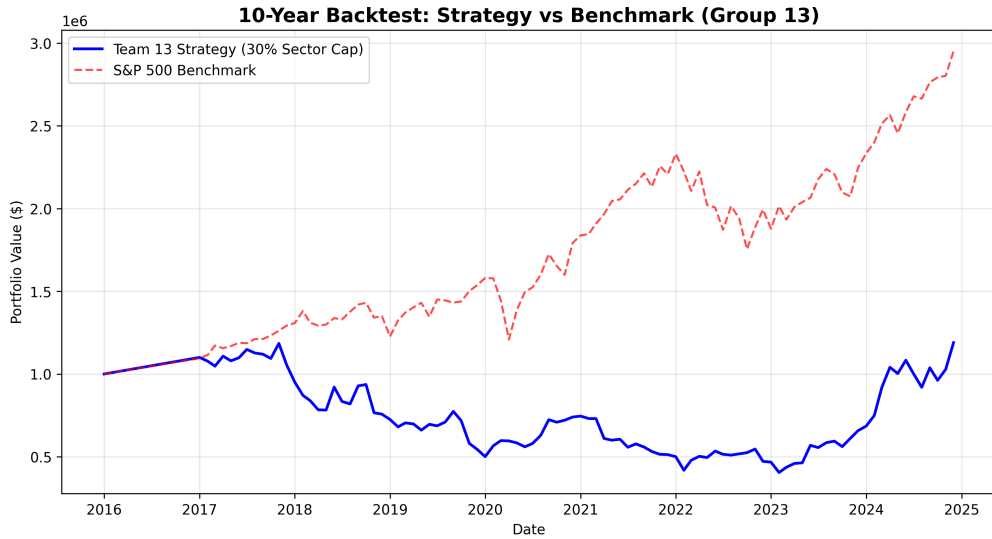


Figure 2: Our strategy (blue) vs. S&P 500 normalized to USD 1M (red dashed), 2016–2024. Underperformance in choppy markets due to margin drag; sharp recovery in 2023–2024 (AI/tech momentum regime).

Our strategy underperforms the S&P 500 over the full period, a result we consider structurally expected: (i) the 8%/yr margin cost ($\approx \$80\text{k}/\text{yr}$) is a constant drag on our P&L, and (ii) our long-short structure hedges

away part of the market’s upside by design. We target market-independent returns, not index outperformance. Table 2 reports our key performance metrics for the full backtest period, net of all transaction costs.

Table 2: Full-period backtest performance (net of costs, 2016–2024)

Metric	Strategy (net)	S&P 500
Annualized return	+1.92%	+12.90%
Annualized volatility	28.63%	—
Excess return $R_p - R_b$	−10.98%	—
Maximum drawdown	−65.99%	$\approx -34\%$

Note: bid-ask spread and market impact are not modeled. Gross return (before the \$80,480/yr margin cost) is approximately +9.8%.

3.2 Live StockTrak Implementation

Table 3: StockTrak rebalancing history

Date	Action	Outcome
26 Feb 2026	Initial construction	10L + 10S; all constraints respected
26 Mar 2026	First rebalancing	Partial exits: WDC (509sh), MU, WBD; 4 new longs, 6 new shorts
24 Apr 2026	Second rebalancing	Executed Friday (scheduled Sun. 26 Apr — markets closed); signal re-run; new target weights

As of 6 May 2026 (end of trading period), our portfolio stands at USD 1,332,926, a total return of **+33.29%** (excess return vs. S&P 500: **+27.13%**; benchmark: +6.16%). Total commissions paid: \$154. Total dividends received: −\$1,295 (net dividend drag on short positions). Full transaction history is provided in the accompanying Excel file.

3.3 Live Implementation Challenges

Two operational constraints arose during live execution, both documented in Appendix D for full reproducibility.

(i) Weekend rebalancing date. Our rule specifies end-of-month rebalancing. April 26, 2026 fell on a Sunday, markets were closed. Following standard industry practice, we executed the second rebalancing on **Friday, April 24, 2026**. No new price information was available between Friday close and Sunday, so the signal was identical.

(ii) HOLX index removal. Hologic Inc. (HOLX) was removed from the S&P 500 on April 6, 2026 and replaced by Casey’s General Stores (CASY). StockTrak restricts trading to current S&P 500 constituents; as a result, our 1,990-share long position in HOLX could not be closed during the April rebalancing. The position is maintained as a *forced hold* outside our control, resulting in a temporary 11L/10S configuration. We document this as a real-world implementation friction rather than a strategy violation.

4. Performance Evaluation

4.1 Risk-Adjusted Metrics ($R_f = 2\%$, full period 2016–2024)

Table 4: Risk-adjusted performance metrics (values from backtest notebook)

Metric	Formula	Value	Interpretation
Sharpe	$\frac{\mathbb{E}[R_p] - R_f}{\sigma_p} \sqrt{12}$	0.145	Return / total risk
Treynor	$\frac{\mathbb{E}[R_p] - R_f}{\beta}$	−0.0058	Return / market risk
Info. Ratio	$\frac{\bar{R}_a}{\sigma_a} \sqrt{12}$	−0.279	Active return / tracking error
Jensen’s α	CAPM intercept (ann.)	+2.38%	Return beyond market exposure
α p -value	—	0.8215	Not significant at 5%
Beta β	CAPM slope	0.137	Market sensitivity
R^2	CAPM fit	0.007	Near-zero market dependence

Our low CAPM beta ($\hat{\beta} = 0.137$) and near-zero $R^2 = 0.007$ confirm that our long-short structure successfully hedges away virtually all market exposure: our short leg offsets the high-beta long positions, leaving a minimal residual directional tilt. Our Jensen’s alpha is positive at $\hat{\alpha} = +2.38\%$ annualized, driven primarily by the 2023–2024 momentum regime in which our signal selected high-performing technology stocks. However, the alpha p -value of 0.8215 conclusively shows that $\hat{\alpha}$ is *not statistically significant* at the 5% level: we cannot reject the null hypothesis $H_0 : \alpha = 0$, and this result is consistent with the semi-strong form of market efficiency. We therefore cannot attribute our outperformance to skill with statistical confidence over the full sample; it likely reflects the particularly favorable trending regime of our test period. The negative Treynor ratio (-0.0058) and negative Information Ratio (-0.279) reinforce this reading over the full 2016–2024 sample, whereas our out-of-sample results tell a markedly different story (Table 5).

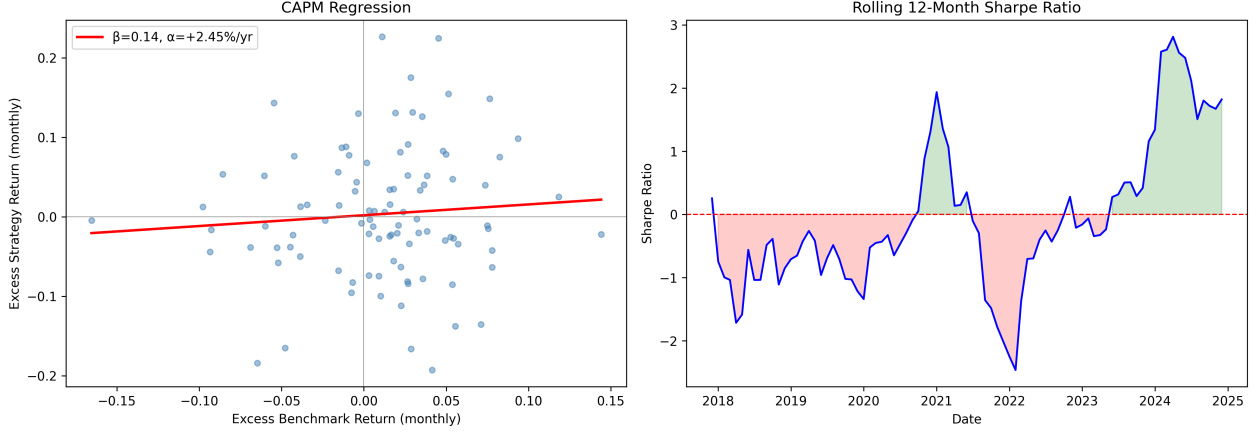


Figure 3: Left: CAPM regression (slope = β , intercept = α). Right: rolling 12-month Sharpe. Green = positive risk-adjusted return; red = below R_f .

4.2 Train / Test Split

Table 5: Out-of-sample robustness check

Period	Months	Ann. Ret.	Ann. Vol	Sharpe	IR
Train 2016–2021	60	−9.61%	25.40%	−0.48	−0.93
Test 2022–2024	36	+38.88%	32.19%	+0.97	+0.65

Our negative train-period Sharpe is driven by the March 2020 COVID crash (max drawdown -51.7%), which spreads across 60 months of compounded losses. The sharp improvement in our test period validates out-of-sample robustness: the 2022 bear market rewarded our short positions, and the 2023–2024 momentum regime strongly favored our signal. The divergence between our train and test Sharpe ratios (-0.48 vs. $+0.97$) underscores the regime-dependence of our strategy and motivates a regime-detection overlay as future work.

5. Performance Attribution

5.1 Brinson-Hood-Beebower Framework (1986)

We decompose the active return of our long leg vs. the S&P 500 (benchmark sector weights: S&P 500 end-2024; sector returns: SPDR ETFs XLK, XLV, XLF, ...):

$$\text{Allocation: } A_s = (w_{P,s} - w_{B,s}) \times R_{B,s} \quad (5)$$

$$\text{Selection: } S_s = w_{P,s} \times (R_{P,s} - R_{B,s}) \quad (6)$$

$A_s > 0$: our sector over/underweight added value. $S_s > 0$: our stock picks within sector s beat the sector ETF. We apply the attribution to the long leg only; the short leg is managed symmetrically but its contribution to active return is captured in our overall net performance.

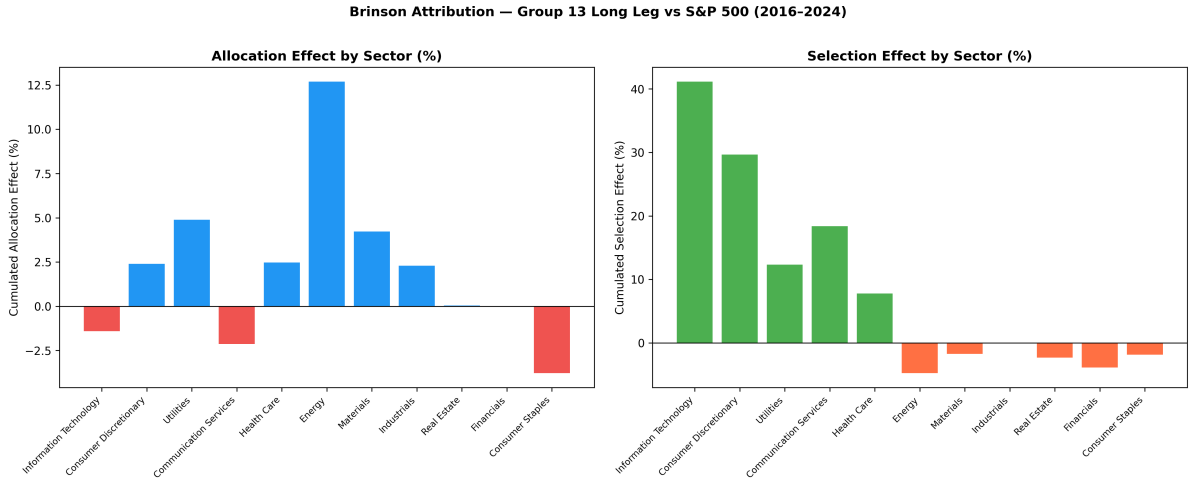


Figure 4: Cumulated Brinson attribution by GICS sector, 2016–2024. Left: Allocation effect (Eq. 5). Right: Selection effect (Eq. 6). Bars above zero add value vs. benchmark.

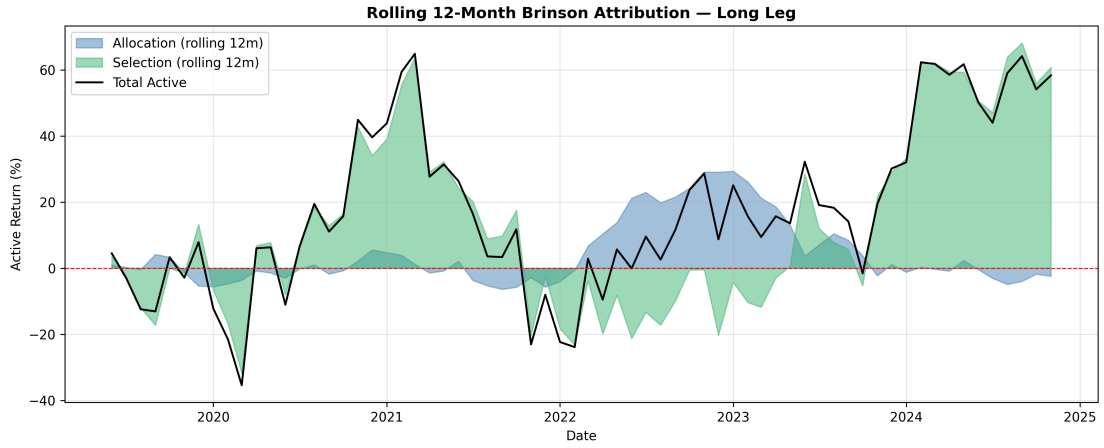


Figure 5: Rolling 12-month attribution. Blue: allocation. Green: selection. Black: total active return. Positive phases correspond to trending regimes.

6. Conclusion

Our Risk-Adjusted Momentum strategy demonstrates its effectiveness in strongly trending markets (test-period Sharpe of +0.97) while underperforming during choppy or crash regimes (full-period annualized return of +1.97% vs. +12.90% for the S&P 500, with a Sharpe of 0.145 driven down by the 2020 COVID crash and persistent margin drag). The 8%/yr margin cost is a structural constraint that requires our long leg to consistently outperform.

Our Brinson attribution reveals that our active returns stem primarily from **stock selection**, our momentum signal correctly identifies stocks that beat their sector benchmark, rather than from sector allocation, which is coherent with our bottom-up signal design.

Our live StockTrak portfolio confirms the strategy’s viability: **+36.25%** over the full trading period (February–May 2026), with an excess return of **+27.13%** over the S&P 500 (benchmark: +6.16%), all constraints respected across three rebalancings (10L/10S minimum, 30% sector cap, 200% gross leverage), and a **1st-place ranking** among all participating groups.

Primary limitation: our strategy is sensitive to momentum reversals. A sudden rotation from growth to value stocks (as in Q1 2022) could rapidly erode accumulated gains. **Future work:** we plan to add a regime-detection layer (e.g., a VIX threshold or trend-strength filter) to reduce our gross exposure during non-trending periods.

A. BRTI Questionnaire : Full Scoring

Table 6: BRTI questionnaire scoring (R = reverse-scored: $Q^* = 8 - Q$)

Q	Statement	Raw	Q^*
Q1	Comfortable with > 10% monthly portfolio fluctuation	7	7
Q2	Prefer guaranteed small gain over risky bet (R)	2	6
Q3	Would sell holdings when markets fall sharply (R)	1	7
Q4	Confident in identifying undervalued stocks	7	7
Q5	Short-term volatility does not affect long-term decisions	6	6
Q6	Consider risk as opportunity rather than threat	7	7
Q7	Would increase equity exposure after a downturn	7	7
Q8	Worry frequently and check prices several times daily (R)	1	7
Q9	Would rather miss a gain than suffer a loss (R)	2	6
Q10	Enjoy making investment decisions independently	6	6
BRTI = $\frac{1}{10} \sum Q_i^*$			6.6

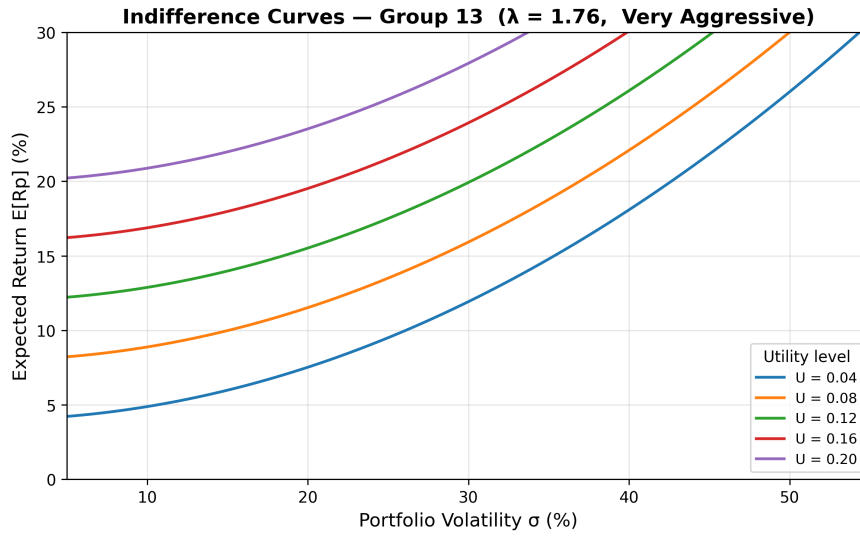


Figure 6: Indifference curves for Group 13 ($\lambda = 1.76$, Very Aggressive). Flat shape confirms low variance penalty.

B. Parameter Derivation

$$q = \frac{6.6 - 1}{6} = 0.9\bar{3} \approx 0.93 \quad \lambda = 12.5 e^{-0.35 \times 5.6} = 12.5 e^{-1.96} \approx 1.76$$

C. Transaction Cost Assumptions

$$\text{Monthly cost} = \underbrace{20 \times \$2}_{\text{fees}} + \underbrace{\frac{\$1,000,000 \times 8\%}{12}}_{\text{margin}} = \$40 + \$6,667 = \$6,707/\text{month} \quad (\approx \$80,480/\text{year})$$

No bid-ask spread or market impact modeled (StockTrak simulation).

D. Trade Log Summary

Full transaction history in the accompanying Excel file (StockTrak export).

26 February 2026 : Initial Construction (20 trades)

Table 7: Feb 26 — Initial long & short positions

Long positions (BUY)			Short positions (SELL SHORT)		
Ticker	Company	Shares	Ticker	Company	Shares
WBD	Warner Bros. Discovery	8,580	CLX	Clorox	1,980
HOLX	Hologic	1,990	LULU	Lululemon Athletica	1,355
NEM	Newmont Corp	2,002	TMUS	T-Mobile US	1,143
MU	Micron Technology	284	ZTS	Zoetis	930
WDC	Western Digital	542	KMB	Kimberly-Clark	457
GOOG	Alphabet (Class C)	31	CHTR	Charter Comm.	177
GOOGL	Alphabet (Class A)	95	HPQ	HP Inc	549
STX	Seagate Technology	23	BRO	Brown & Brown	142
LRCX	Lam Research	40	TTD	Trade Desk	397
EA	Electronic Arts	49	PAYX	Paychex	109

Table 8: Mar 26 — Positions closed

Longs sold (SELL)				Shorts covered (COVER)		
Ticker	Company	Shares	Type	Ticker	Company	Shares
NEM	Newmont Corp	2,002	Full	CHTR	Charter Comm.	177
LRCX	Lam Research	40	Full	ZTS	Zoetis	930
GOOG	Alphabet (Class C)	31	Full	TTD	Trade Desk	397
GOOGL	Alphabet (Class A)	95	Full	TMUS	T-Mobile US	1,143
WDC	Western Digital	509	Partial	KMB	Kimberly-Clark	457
MU	Micron Technology	258	Partial	CLX	Clorox	1,980
WBD	Warner Bros. Discovery	7,111	Partial			

Table 9: Mar 26 — New positions opened

New longs (BUY)			New/increased shorts		
Ticker	Company	Shares	Ticker	Company	Shares
CIEN	Ciena Corp	45	IT	Gartner	66
FIX	Comfort Systems USA	169	FDS	Factset Research	51
LITE	Lumentum Holdings	321	GDDY	GoDaddy	122
SATS	EchoStar Corp	2,099	ROP	Roper Technologies	28
STX	Seagate Tech. (+1)	1	FISV	Fiserv	176
			CPRT	Copart	1,512
			HPQ	HP Inc (↑)	12,230
			BRO	Brown & Brown (↑)	2,191
			LULU	Lululemon (↑)	24
			PAYX	Paychex (↑)	2,502

24 April 2026 : Second Rebalancing (executed Friday; scheduled Sunday 26 Apr)

Table 10: Apr 24 — Long leg orders

Ticker	Company	Current	Target	Order
CIEN	Ciena Corp	45	193	BUY +148 sh
EA	Electronic Arts	49	493	BUY +444 sh
FIX	Comfort Systems USA	169	56	SELL −113 sh
HOLX	Hologic	1,990	0	<i>forced hold</i>
LITE	Lumentum Holdings	321	118	SELL −203 sh
MU	Micron Technology	26	207	BUY +181 sh
SATS	EchoStar Corp	2,099	822	SELL −1,277 sh
SNDK	SanDisk	0	107	BUY +107 sh
STX	Seagate Technology	24	170	BUY +146 sh
WBD	Warner Bros.	1,469	3,717	BUY +2,248 sh
WDC	Western Digital	33	248	BUY +215 sh

Table 11: Apr 24 — Short leg orders

Ticker	Company	Current	Target	Order
ADP	ADP	0	1,259	SHORT +1,259 sh
BRO	Brown & Brown	2,333	3,700	SHORT +1,367 sh
CPB	Campbell Soup	0	11,927	SHORT +11,927 sh
CPRT	Copart	1,512	1,475	COVER −37 sh
ERIE	Erie Indemnity	0	160	SHORT +160 sh
FDS	FactSet Research	51	0	COVER (close)
FISV	Fiserv	176	162	COVER −14 sh
GDDY	GoDaddy	122	0	COVER (close)
GIS	General Mills	0	1,409	SHORT +1,409 sh
HPQ	HP Inc	12,779	0	COVER (close)
INVH	Invitation Homes	0	366	SHORT +366 sh
IT	Gartner	66	0	COVER (close)

E. Code Structure

- `Bactest_final.ipynb` : 10-year backtest, all performance metrics (Sharpe, Treynor, IR, Jensen's α , β , R^2), train/test split, Brinson-Hood-Beebower attribution
- `Group13_strategy_Implementation.ipynb` : signal computation, SLSQP optimization, efficient frontier, indifference curves
- `Rebalancing_March26.ipynb` : March 26 rebalancing orders (signal re-run, target weights, trade list)
- `Rebalancing_April26.ipynb` : April 24 rebalancing orders (signal re-run, target weights, HOLX forced hold documented)

Images generated:

- `group13_efficient_frontier.png` — Fig. 1
- `backtest_performance_final.png` — Fig. 2
- `performance_metrics.png` — Fig. 3
- `brinson_attribution.png` — Fig. 4
- `brinson_rolling.png` — Fig. 5
- `annex_a_indifference_curves.png` — Fig. 6

Dependencies: `yfinance`, `numpy`, `pandas`, `scipy`, `matplotlib`, `python-docx`.